



FITSYNC: AI AND DT SOLUTION FOR MODERN GYMS

What does Fitsync aim at ?

~ Fitsync is a novel approach to integrate the many ideas from the existing technologies and world. Here in Fitsync, we provide a never before attempted solution to tackle one of the most critically relevant social problem - increase in the number of deaths of people during workouts. This has been becoming a trending topic because of which gym and workout enthusiasts need to be more self-assured and must have the opportunity to self-assess their own health conditions. They need to be guaranteed safety of their health. Recent news has been reporting sudden deaths due to cardiac arrests and overexertions. No matter how much physical gyms and physical trainers can provide nutritious insights and healthy diets, it cannot give the assurance nor credibility of a real-time monitoring system of that person working out, where the real-time system is capable of recording past-data and giving insights on the present data based on the past data of that person.

Fitsync presents before you an AI system comprising of ML model trained on ML algorithms based on fitness datasets and delivers to you your personal workout status. In Fitsync, you can view graphs on your live heart rates,

predicted heart rates determined thoroughly by studying patterns by the trained model, and also the anomaly detection graph part.

Fitsync approaches to implement this real time heart rate monitoring through the cutting edge and currently rising technology Digital Twin. Numerous researches are going on in the names of Digital Twin, but very few have taken up the courage to implement it. And that too rare numbers are in Fitness and Medical Sector when it comes to Digital Twin.

Fitsync has only good deeds. It is designed for the health conscious community, for them to safely workout without being misunderstood or misleading. Fitsync is designed to help the exercise enthusiasts get a better visual representation of their stats. Here we have only focused on heart rates as a factor to monitor because this is a never before implemented approach with Digital Twin. We've created the Digital Twin model in MATLAB MATHWORKS SIMULINK model and it was a successful attempt. As for the live input heart rate reading, we've programmed with arduino and a bpm sensor. The web interface set up using node.js acts as the primary contact between the user and the Digital Twin. Apart from that, there is a login page, signup page and signout sessions for the user, database connected through mongodb.

256 batch size, 5 epochs

Deep Learning ToolBox